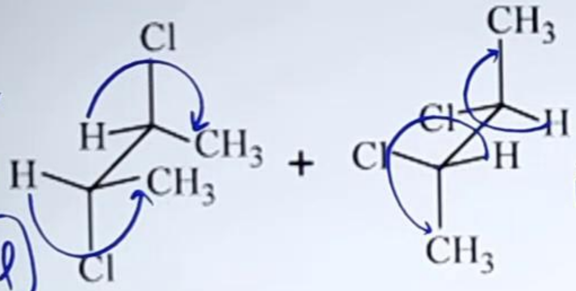
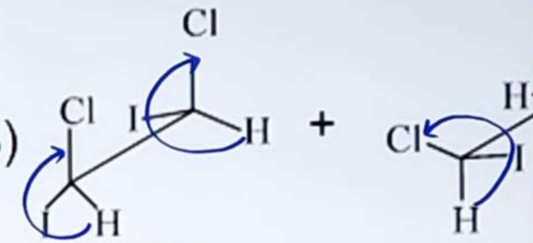
09:55

Which of the following pairs of compounds are identical?

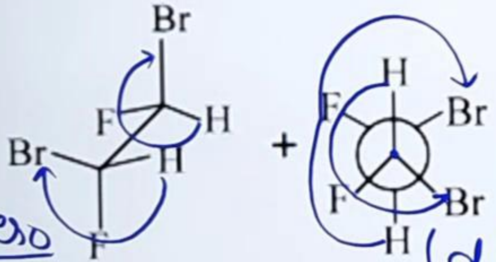
✓ (A)  identical

d|l

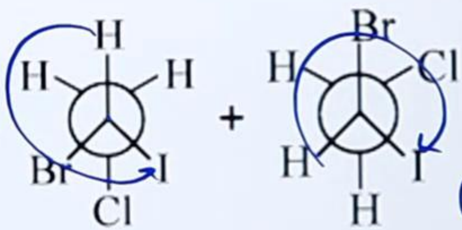
× (B) 

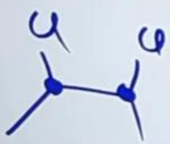
(meso) (d/l)

(diastereomers)

× (C)  (diastereomers)

meso (d,l)

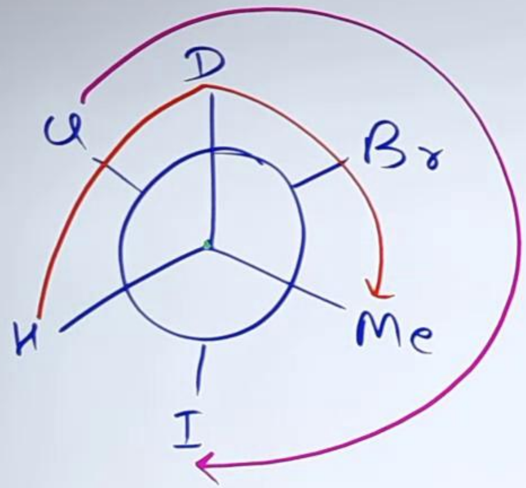
× (D)  enantiomers

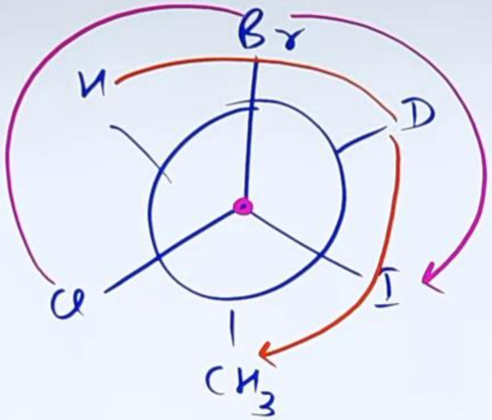


d, l, meso

Ans = A

Q



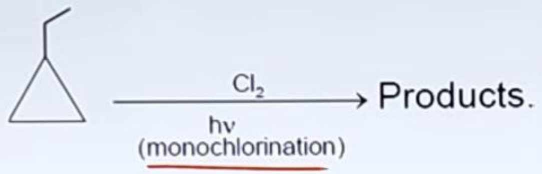


Enantiomers

10:07

10:13

Answer the following questions based on given reaction.



(a) Total number of theoratically products (including stereo) are

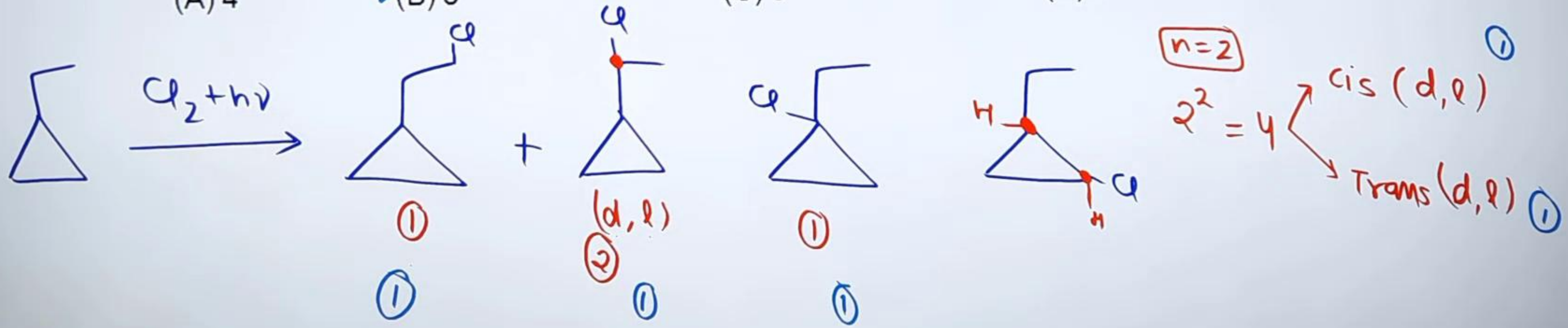
(A) 4 (B) 6 (C) 8 (D) 10

(b) How many products are resolvable.

(A) 2 (B) 4 (C) 5 (D) 6

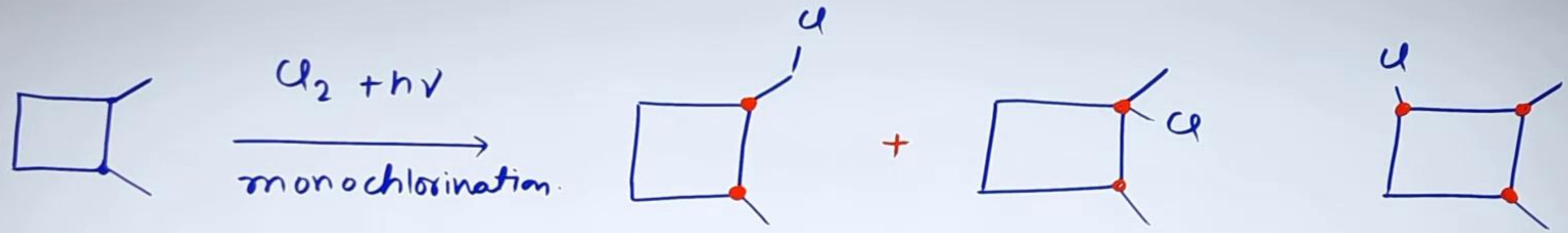
(c) How many fractions are present on fractional distillation?

(A) 4 (B) 5 (C) 6 (D) 8



10:20

Q



all stereo isomers

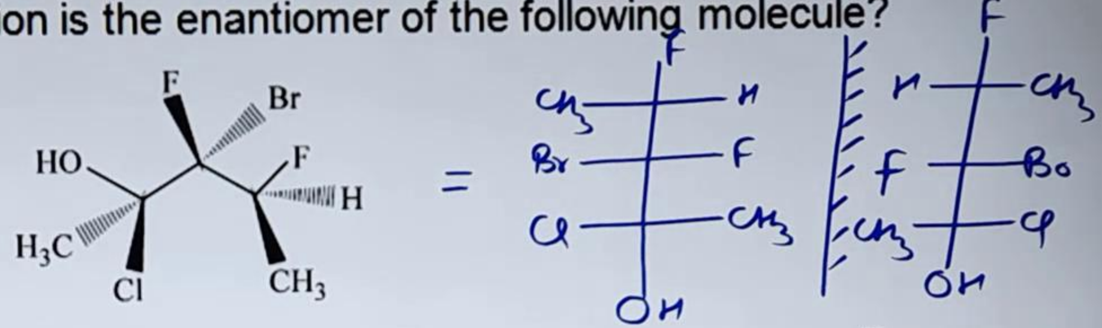
$n = 2$	$n = 2$	$n = 3$
$2^n = 4$	$2^n = 4$	$2^3 = 8$
cis (d, l), Trans (d, l)	cis (d, l), Trans (d, l)	Pair of enantiomers (4)

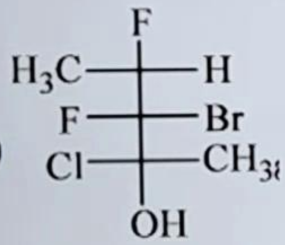
$\Rightarrow$ Total product $\Rightarrow$ 16

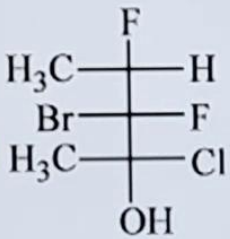
$\Rightarrow$ no. of fractions on fraction dis.... (8)

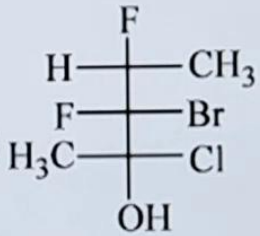
10:30

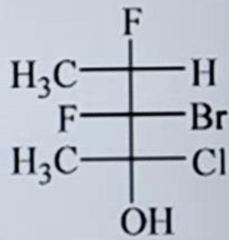
Q. Which of the following Fischer projection is the enantiomer of the following molecule?

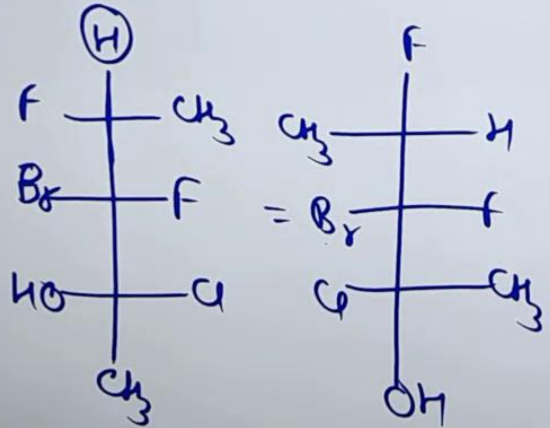


(A) 

(B) 

(C) 

(D) 

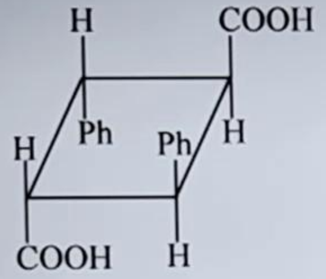


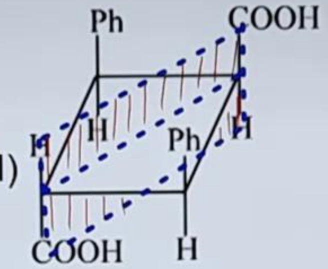
$\boxed{\text{Ans} = C}$

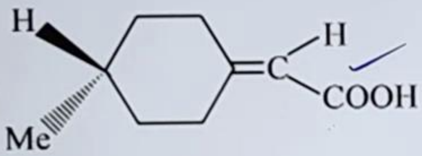
10:37

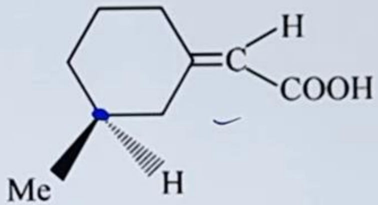
Q

Match the column.

(I) 

(II) 

(III) 

(IV) 

Column I		Column II	
(A) Plane of symmetry	Q	(P)	I
(B) Centre of symmetry	(P)	(Q)	II
(C) Show geometrical isomerism	(PQS)	(R)	III
(D) Show optical isomerism	(R, S)	(S)	IV

10:41

Q

Dextrorotatory α–pinene has a specific rotation $[\alpha]_D^{20} = +51.3°$. A sample of α–pinene containing both the enantiomers was found to have a specific rotation value $[\alpha]_D^{20} = +30.8°$. The percentages of the (+) and (–) enantiomers present in the sample are, respectively.

(A) 70% and 30% (B) 80% and 20% (C) 20% and 80% (D) 60% and 40%

Sol.

$(d + l)_{mix}$ $[\alpha]_{mix.} = +30.8°$ $\boxed{d > l}$

Pure d $[\alpha] = 51.3°$

optical purity (enantiomeric excess) $= \dfrac{[\alpha]_{mix}}{[\alpha]_{pure}} \times 100 = \dfrac{30.8°}{51.3°} \times 100 = 60.03\% \simeq 60\%$

$\Rightarrow d - l = 60$

$d + l = 100$

$\boxed{\text{Ans} = B}$

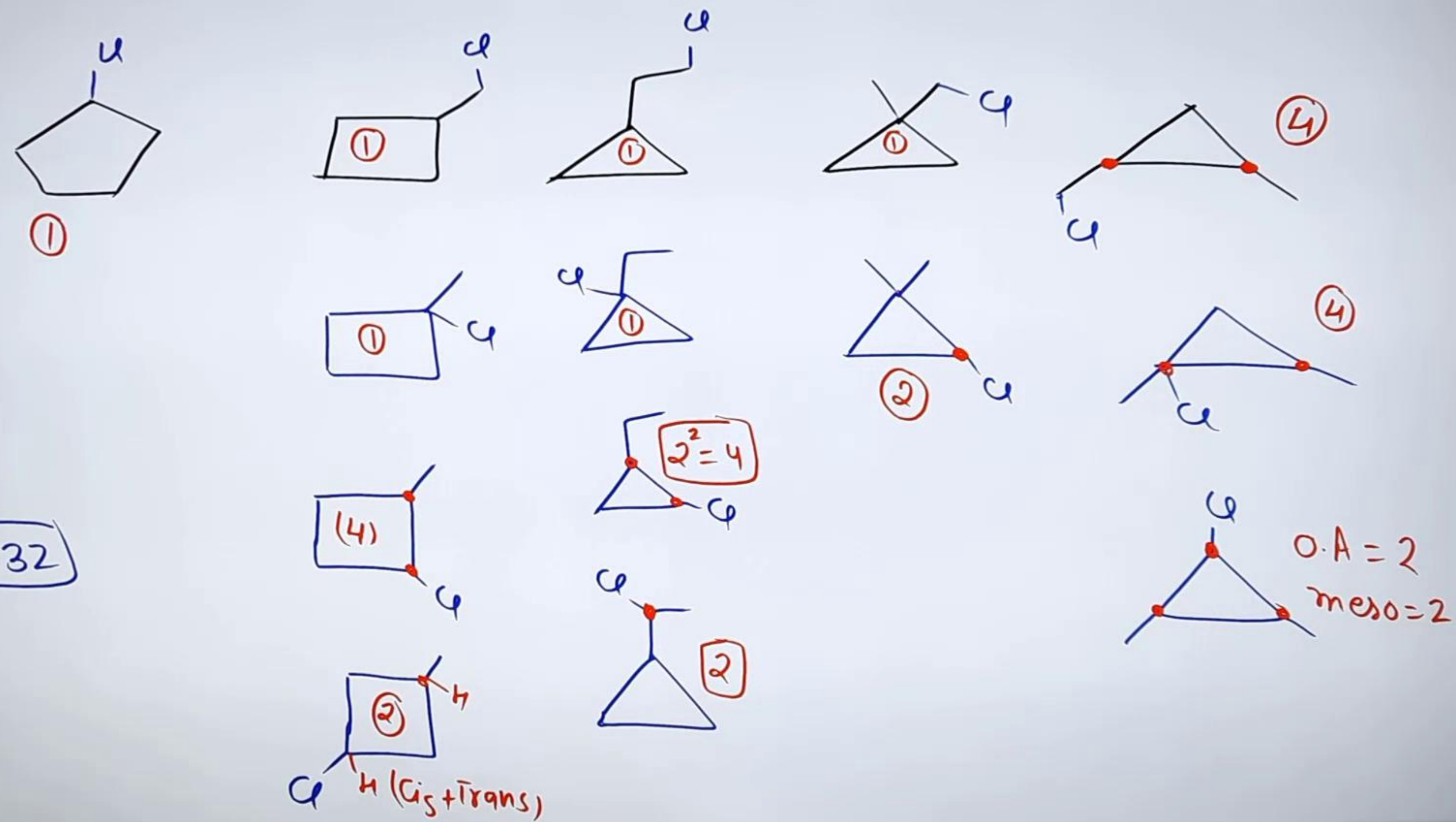
competishun

Power of real gurus

10:57

Q. Identify all possible cyclic isomers of C_5H_9Cl $\boxed{DU = 1}$

Sol

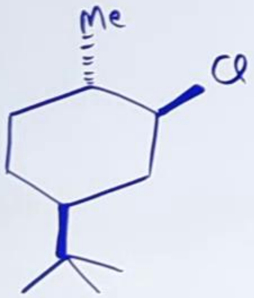


$\boxed{Ans = 32}$

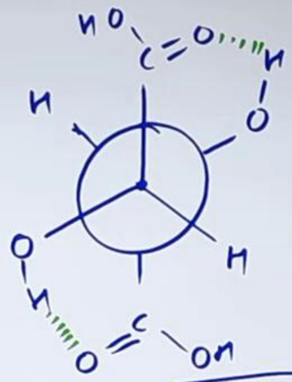
Q Draw most stable conformation -

1. meso tartaric acid

2.



1.



Anti conformation most stable

$\boxed{\mu \neq 0}$

2.